

JIVESH ARORA

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EDUCATION

University of Waterloo, Canada

Jan 2025 - Aug 2029 (Expected)

Bachelor of Computer Science (Honours), Co-op

Relevant Coursework: Data Structures and Algorithms, Statistics for CS, Databases, Logic and Computation

EXPERIENCE

Royal Bank of Canada (RBC)

January 2026 - August 2026

Insight Analyst, Digital Analytics (Personal & Commercial Banking)

Toronto, Canada

- Built a daily PySpark → Hive data pipeline orchestrated with YAML-defined Airflow DAGs, cutting campaign reporting from ~8 hours to ~10 minutes per day across 3+ Tableau dashboards.
- Diagnosed a Tableau fan-out bug inflating an iPad campaign metric ~27x; fixed with FIXED LOD expressions.
- Modelled sales, postal-code, and campaign data to track new chequing accounts and response by region.
- Migrated legacy QuickSight reporting to Tableau, converting SQL logic to PySpark and shipping through Git.

Computing Research Association (CRA), University of Waterloo

September 2026 - Present

AI/ML Undergraduate Researcher, UR2PhD Program (Prof. Marina Meila)

Waterloo, Canada

- Selected for CRA's UR2PhD research program to study when dimensionality-reduction methods (t-SNE, UMAP, spectral embedding) preserve or distort the true shape of high-dimensional data.
- Prototyping an automatic distortion score in Python on megaman's local metric estimates, replacing the group's visual inspection of distortion ellipses, toward a proposal using synthetic manifolds (swiss roll, torus).

Off-Campus Don, WUSA

August 2025 - April 2026

- Supported a community of 5,000+ off-campus students with guidance on housing, tenancy, and safety.

PROJECTS

Wanderers | [Site](#) |  [Source Code](#)

Team of 3 | Pitched at Velocity Pitch Competition, Sept 2026

Tech Stack: React Native (Expo), TypeScript, Next.js, React, Supabase (Postgres/Auth), Tailwind

- Built the React Native (Expo) client for a social platform connecting University of Waterloo students: Expo Router navigation, authentication, discovery, messaging and map screens with shared auth contexts.
- Engineered a two-step authentication flow with Supabase Auth using password verification followed by emailed OTP; implemented signup, verification, and password-reset routes.

Cold-Start Recommendation System | [Site](#) |  [Source Code](#)

Tech Stack: Python, PyTorch, Sentence-Transformers, FastAPI, Docker, AWS SageMaker, Streamlit

- Built a hybrid recommender over 800K+ Amazon reviews, combining PyTorch matrix factorization with MiniLM sentence embeddings through confidence-weighted fusion, improving Recall@10 by 1.17pp.
- Caught a lookahead leak (implausible 86.7% score) and a fusion scale mismatch via leave-one-out validation.
- Deployed as a Dockerized FastAPI service on AWS SageMaker; debugged an instance-only OOM crash traced to silent float64 promotion and fixed it with explicit float32 casts.

OCC Community Assistant | [Site](#) |  [Source Code](#)

Tech Stack: Node.js, Express.js, React, SQLite, Groq API (GPT-OSS), BM25

- Built a hand-wired agent pipeline on Groq for WUSA's off-campus community: an LLM intent router, BM25 retrieval over 125 official sources and 42 FAQs, and a 4-step tool-calling loop for tickets and escalation.
- Wrote a rule-based safety critic (not a second LLM) that force-escalates crisis cases, logging decisions to SQLite.

SKILLS

Languages & Tools

Python, C, C++, TypeScript, JavaScript, SQL, Git, Linux

Web & Backend

React, Next.js, React Native (Expo), Node.js, Express.js, FastAPI, REST APIs

Data & Cloud

PostgreSQL (Supabase), MongoDB, SQLite, Docker, PySpark, Hive, Airflow, AWS

AI/ML

PyTorch, Sentence-Transformers, LLM agents, function calling, RAG, BM25 retrieval

Certifications

AWS Certified Cloud Practitioner (CLF-C02), AWS Certified AI Practitioner (AIF-C01)